

EE150 Signal and System

Homework 8

Note:

- Please provide enough calculation process to get full marks.
- Please submit your homework to Gradescope in PDF version.
- It's highly recommended to write every exercise on a single sheet of page.
- Late submissions will have points deducted according to the penalty policy.
- Please use English only to complete the assignment, solutions in Chinese are deducted according to the penalty policy.
- Plagiarizer will get zero points.
- The full score of this assignment is 100 points.

Problem 1. (15pts)

Use the definition of the Z-transform to find the Z-transform expression and the corresponding region of convergence:

(a)

$$x[n] = -\left(\frac{1}{5}\right)^n u[-n - 1]$$

(b)

$$x[n] = \left(\frac{1}{3}\right)^n \{u[n] - u[n - 11]\}$$

(c)

$$x[n] = \left(\frac{1}{5}\right)^n \cos\left[\frac{\pi}{3}n + \frac{\pi}{4}\right] u[n]$$

Problem 2. (15pts)

Answer the questions using the properties of the Z transform below:

(a) Determine the inverse Z transform of $X(z)$

$$X(z) = \frac{3z^{-4}}{(1 - \frac{1}{5}z^{-1})^2}, |z| > \frac{1}{5}$$

(b) Determine the Z transform of $y[n]$

$$y[n] = 3^n \sum_{k=-\infty}^n \left\{ \left(\frac{1}{2}\right)^k u[k] \right\}$$

Problem 3. (20pts)

Consider a LTI system whose zero-pole plot is shown below, and the system function $H(z)$ is rational, with the impulse response function defined as $h[n]$:

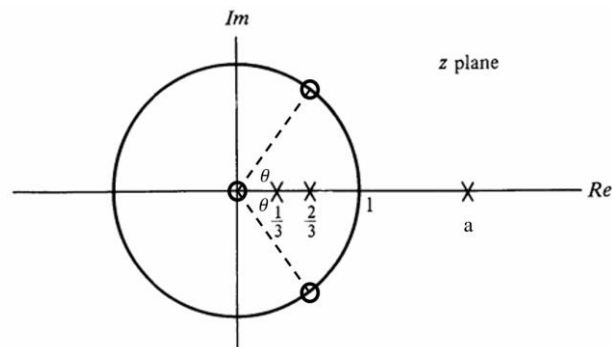


Figure 1. Zero-pole plot for problem 3

- a. Determine whether the following statements are true or false and explain your judgment.
 1. If the system is stable, then it is causal.
 2. $H(e^{j\omega})$ could not be zero for any ω .
 3. $h[n]$ is finite duration.
 4. If the system is stable, then $F\{(\frac{5}{4})^n h[n]\}$ converges.
- b. Find out $H(z)$ if we know:
 - $H(z)$ has exactly three poles and three zeros.
 - $H(z)$'s ROC is $|z| > 3$
 - $h[0] = 3$
 - $\theta = \frac{\pi}{3}$

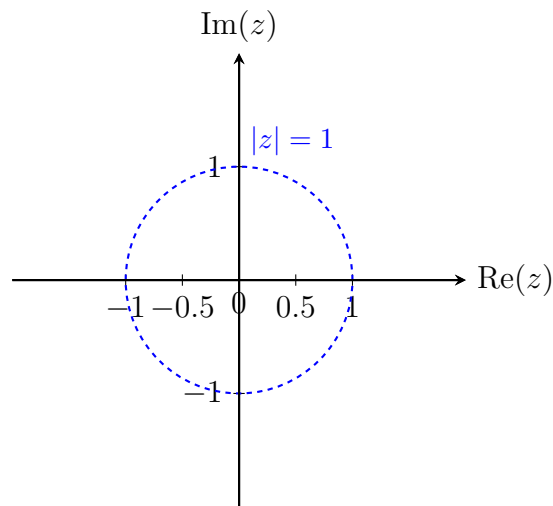
Exercise 4. (15pt)

Consider a stable first-order discrete-time LTI system with a rational system function $H(z)$. The pole-zero plot of the system function $H(z)$ is given as follows:

- There is one zero at $z = 0$;
- There is one pole at $z = \frac{1}{2}$;
- When the input of the system is given by $x[n] = e^{j2\pi n}$, the corresponding output of the system is $y[n] = \frac{12}{5}e^{j2\pi n}$

Determine:

1. The system function $H(z)$, and draw the corresponding ROC.
2. Using the geometric evaluation method based on the pole-zero diagram, calculate the magnitude and phase responses at $\omega = 0$ and $\omega = \frac{\pi}{3}$.



Exercise 5. (15pt)

Consider a discrete-time LTI system described by the difference equation:

$$y[n] - 0.5y[n-1] - 0.5y[n-2] = x[n-1]$$

Given $y[-1] = 0$, $y[-2] = 2$, when $x[n] = \left(\frac{1}{2}\right)^n u[n]$, determine the zero-state response, zero-input response, and the total response of the system.

Exercise 6. (15pt)

Given a stable discrete-time LTI system with input signal $x[n]$ and output signal $y[n]$, satisfying the following difference equation:

$$y[n] + c_1 y[n-1] + c_2 y[n-2] = x[n] + d_1 x[n-1] + d_2 x[n-2]$$

where c_1, c_2, d_1, d_2 are constants. The system has one pole located at $\frac{1}{2}$, a second-order zero located at $\frac{1}{3}$, and the system gain at zero frequency (which is $H(1)$) is $\frac{32}{27}$.

1. Find the values of c_1 , c_2 , d_1 , and d_2 .
2. Draw the direct form block diagram of the system.